# Daniel Hardesty Lewis

*Design Engineer, Analytical Interfaces and AI Workflows*

**Hiring Team**
*GSMC*

February 6, 2026

Dear Hiring Team,

I am applying for the Design Engineer, Frontend and Human AI Interaction role. I build high performance analytical interfaces where the user experience is the product, including the workflow, the trust model, the latency budget, and the human decision loop. I do best work when an interface must turn complex quantitative analysis into fast, legible, and inspectable workflows that improve discretionary decisions.

I have shipped end to end systems that match your mission. At Properlytic, I built and deployed interactive uncertainty visualization using fan charts for a diffusion based forecasting system, backed by leakage safe expanding window evaluation and delivered through a live web prototype at houston-price-prediction.vercel.app. The interface required careful design around uncertainty and model selection tradeoffs, and I am extending it with explanation and evidence views that let a user inspect inputs, sensitivity, and failure modes. That work draws on my industry sponsored research at Columbia, where I derived tractable per feature attribution methods designed for stability across mutual fund market regimes and used them to debug regime dependent model behavior.

At PoliBOM, I specified agentic workflows and data schemas for bill of materials parsing, retrieval, and tariff simulation using Airflow, embeddings, FastAPI, Supabase, and Redis, and I drove development of a conversational interface that compresses multi step compliance analysis into a single decision surface. At Summit Geospatial and previously at TACC, I shipped production data delivery systems where latency, uptime, and provenance were user facing constraints, so stability and observability were part of the product.

In this role, I would prioritize three outcomes. I would build and maintain high performance visualization of analysis that supports discretionary trading research workflows. I would help shape new workflows that integrate Claude and LLM reasoning loops into the research process in a way that is usable and trustworthy. I would deliver production grade frontend systems with strong ownership, emphasizing stability, predictable performance, and reliable operation.

Your focus on the physical world manufacturing economy fits my recent tariffs and supply chain decision support work at PoliBOM. Your focus also fits my prior safety critical hazard decision support work at TACC during live disasters, where turnaround time and reliability affect public safety and can have life safety consequences. I would welcome the opportunity to walk through my shipped interfaces and the design decisions behind them, including model architecture choices, leakage safe evaluation, and backtests.

I do not require sponsorship to work in the United States.

Sincerely,

**Daniel Hardesty Lewis**

*encl: houston-price-prediction.vercel.app • polibom.com • summitgeospatial.com*

**Daniel Hardesty Lewis**
*500 Riverside Dr – New York, NY 10027 – United States*
*+1 (713) 371-7875* • *daniel@summitgeospatial.com* • *dlewis.ai* • *dhardestylewis*
*dhardestylewis* • *Google Scholar*

# Daniel Hardesty Lewis

*Design Engineer, Analytical Interfaces and AI Workflows*

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

## Core Expertise

8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

## Technologies

| | |
|---|---|
| Modeling | PyTorch; TensorFlow; scikit-learn; Hugging Face Transformers; LoRA/QLoRA/PEFT |
| LLM apps | embeddings; RAG; LangChain; LangGraph; DSPy; FAISS |
| Data stack | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase |
| Infrastructure | Linux/Unix; Bash; Git; Docker; Kubernetes; Ray; Spark; Dask; Airflow; CI/CD; observability |
| Additional | CUDA; C++; TypeScript; React |

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Properlytic*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation; owned end-to-end workflow from source data to web delivery.

- Deployed an NYC **semi-supervised VAE** current-price model using **tax and sales records** to handle **sparsity and noise**; achieved 12% holdout error (vs. Zillow's 8.4% internal apples-to-apples).
- Shipped a Houston diffusion-based forecasting system with fan charts; achieved as low as ∼8% annualized compounding error, balancing design choices of error vs. trajectory realism, delivered via prototype: houston-price-prediction.vercel.app
- Refined product positioning and execution milestones via strategic feedback from a16z, MetaProp, and others.
- Repo: github.com/dhardestylewis/v0-properlytic-8v

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Directed development of a **conversational AI interface** simplifying global trade workflows; enabled decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with the founder of ScoutyAI and tariff economists at Columbia Business School to validate product-market fit and pricing assumptions.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Developed the web distribution platform end-to-end including data pipeline, tiling, hosting, delivery UX to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Co-founded summitgeospatial.com to commercialize and sustain state-level terrain models developed for the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, executing million-node distributed jobs while managing multi-million-dollar compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Reviewed Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, allocating millions of dollars in computing resources on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

## Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Math benchmarking and finetuning of Deepseek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset conditional variational autoencoder (CVAE) latent factor model with Skew-T Mixture priors; achieved 85% $R^2$ vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

## Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization.

# Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

# Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

# Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

# Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

# Languages

Spanish Native Proficiency

French Professional Working Proficiency

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